CO2-AT2- ASSIGNMENT 1

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QUESTIONS

Q1
Digital Music Streaming Playlist Management System
50 marks

1/1 answered

Q1 Digital Music Streaming Playlist Management System

50 marks

A music streaming company hosts millions of songs uploaded by record labels, artists, and independent creators. Since content is received from multiple sources, duplicate song IDs, inconsistent artist names, repeated album titles, and multiple playlist entries are common. The company plans to build a Java application that organizes playlists efficiently while eliminating duplicate records using the Java Collection Framework.

Develop a Java application that performs the following operations:

a) Read Song ID, Song Title, Artist Name, Album Name, Genre, Duration, Language, and Playlist Name.

b) Use String Handling methods to standardize song titles, artist names, and album names.

c) Store songs using an ArrayList.

d) Remove duplicate Song IDs using a HashSet.

e) Maintain a HashMap mapping Playlist Names with song collections.

f) Display playlist details using Iterator.

g) Implement Generic Collections.

h) Generate genre-wise playlist reports and explain the advantages of Collection Framework classes in multimedia applications.

Your Code

```
import java.util.*;
public class Main{
 public static void main(String[] args){
  Scanner sc=new Scanner(System.in);
  ArrayList<Song> list=new ArrayList<>();
  HashSet<String> ids=new HashSet<>();
  HashMap<String,ArrayList<Song>> playlists=new HashMap<>();
```

Input

```
Pop
3.2
English
Favorites
```

Output

```java
import java.util.*;
public class Main{
 public static void main(String[] args){
  Scanner sc=new Scanner(System.in);
  ArrayList<Song> list=new ArrayList<>();
  HashSet<String> ids=new HashSet<>();
  HashMap<String,ArrayList<Song>> playlists=new HashMap<>();
  System.out.print("Enter number of songs: ");int n=sc.nextInt();sc.nex
  for(int i=0;i<n;i++){
   System.out.println("\nSong "+(i+1));
   System.out.print("Song ID: ");String id=sc.nextLine();

   if(ids.contains(id)){
    System.out.println("Duplicate Song ID!");continue;
   }

   System.out.print("Song Title: ");String title=sc.nextLine();
   System.out.print("Artist Name: ");String artist=sc.nextLine();System
   System.out.print("Genre: ");String genre=sc.nextLine();System.out.pr
   System.out.print("Language: ");String language=sc.nextLine();System.
   Song s=new Song(id,title,artist,album,genre,duration,language,playli
   ids.add(id);list.add(s);

   playlists.putIfAbsent(s.playlist,new ArrayList<>());playlists.get(s.
  }

  System.out.println("\n--- PLAYLIST DETAILS ---");
  Iterator<String> pIt=playlists.keySet().iterator();
  while(pIt.hasNext()){
   String p=pIt.next();System.out.println("Playlist: "+p);
   Iterator<Song> sIt=playlists.get(p).iterator();
   while(sIt.hasNext()){
```

```
S101
Shape of You
Ed Sheeran
Divide
Pop
3.5
English
Favorites
S102
Blinding Lights
The Weeknd
After Hours
Pop
3.2
English
Favorites
```

Output

```
Enter number of songs:
Song 1
Song ID: Song Title: Artist
Name: Album Name: Genre:
Duration: Language: Playlist
Name:
```

```java
  HashMap<String,ArrayList<Song>> genres=new HashMap<>();
  for(Song s:list){
   genres.putIfAbsent(s.genre,new ArrayList<>());genres.get(s.genre).ac
  }

  for(String g:genres.keySet()){
   System.out.println("Genre: "+g);
   for(Song s:genres.get(g)){
    System.out.println("  "+s.title+" by "+s.artist);
   }
  }

  System.out.println("\n--- ADVANTAGES ---");
  System.out.println("1.ArrayList stores songs dynamically.");
  System.out.println("2.HashSet avoids duplicate IDs.");
  System.out.println("3.HashMap stores playlists efficiently.");
  System.out.println("4.Iterator helps traverse collections.");
 }
}
class Song{
 String id,title,artist,album,genre,language,playlist;
 double duration;
 Song(String id,String title,String artist,String album,String genre,do
  this.id=id;this.title=title.trim().toUpperCase();this.artist=artist.t
  this.album=album.trim().toUpperCase();this.genre=genre.trim().toUpper
  this.duration=duration;this.language=language.trim().toUpperCase();th
 }
 public String toString(){
  return id+" "+title+" "+artist+" "+album+" "+genre+" "+duration+" "+l
 }
}
```